Practice Set 14 Paper 2 – Urology

1. Q1. 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.
2. Q2. Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder.
3. Q3a. Role of UDS before prostate surgery
4. Q3b. Zolendronic acid
5. Q3c. LHRH analogue
6. Q3d. Expectant treatment of CA prostate
7. Q3e. Molecular staging of CA prostate
8. Q3f. CRE & BEP

Practice paper compiled from the question bank.

Q1. 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.

Type: LAQ (25 marks) (related: prognosis in 2019-Summer Paper 4, Q3e; staging of non-organ-confined disease in Practice Set 10 Paper 2, Q1)

"Mixed voiding symptoms" is read as mixed storage and voiding LUTS (lower urinary tract symptoms).

Introduction

- In a 48-year-old, a PSA (prostate-specific antigen) of 18 ng/mL is far above what BPH (benign prostatic hyperplasia) usually produces. **Prostate cancer must be excluded first**; the LUTS are then treated on their own merits.
- PSA is organ-specific, not cancer-specific: BPH, prostatitis, UTI (urinary tract infection), retention and instrumentation also raise it. At PSA >10 ng/mL, biopsy yield is 43–65% with a normal DRE (digital rectal examination) and 70–90% with a suspicious one.

History

- **LUTS:** IPSS (International Prostate Symptom Score) with bother score; storage versus voiding; retention, haematuria, recurrent UTI.
- **Red flags:** bone pain, weight loss, leg oedema, cord compression.
- Family history (prostate cancer <65 years; BRCA (breast cancer gene)-related cancers), recent catheter, fever or perineal pain (prostatitis), 5ARI (5α-reductase inhibitor) use, neurological disease, erectile function.

Examination

- **DRE:** size, nodule, induration, extension, fixity (clinical T stage). Palpable bladder, neurological and spinal examination.

Investigations

Test	Purpose
Urinalysis, culture	Exclude UTI or prostatitis; treat and repeat PSA later (no empirical antibiotics without symptoms)
Repeat PSA, free PSA	Confirm the rise; low free PSA favours cancer
Creatinine, ultrasound KUB (kidney, ureter, bladder)	Upper tracts, prostate volume (PSA density)
Uroflowmetry, PVR (post-void residual)	Baseline LUTS assessment
mpMRI (multiparametric MRI) before biopsy	PI-RADS (Prostate Imaging Reporting and Data System) score; guides targeted cores; local staging
Biopsy	Targeted (PI-RADS ≥3) **plus systematic** 12 cores; **transperineal** preferred (lower sepsis)

- At PSA 18, biopsy is needed **even if MRI is negative**. Blood or urine biomarkers (PHI (Prostate Health Index), 4Kscore, PCA3) matter mainly in the 2–10 ng/mL grey zone.

Staging if cancer is found

- Grade group, cores involved, PSA, T stage → risk group. PSA 10–20 makes him at least **intermediate risk** (D'Amico, NCCN (National Comprehensive Cancer Network)).
- **High risk:** bone scan with CT (computed tomography), or **PSMA PET (prostate-specific membrane antigen positron emission tomography)**. Organ-confined disease is <50% at PSA >10; nodes about 20% at PSA >20.

- **Germline testing** (BRCA2, mismatch repair genes) for high-risk or metastatic disease or strong family history.

Management of the cancer (life expectancy >25 years)

Situation	Management
Low or favourable intermediate risk	AS (active surveillance) is possible, but uncommon at PSA 18
Localised intermediate or high risk	**RP (radical prostatectomy)**, usually robotic, with extended PLND (pelvic lymph node dissection) when node risk exceeds the nomogram threshold; or **EBRT (external beam radiotherapy) + ADT (androgen deprivation therapy)**: 4–6 months intermediate, 18–36 months high risk
Locally advanced (T3–T4, N1)	Multimodal: RP ± adjuvant or early salvage RT, or EBRT + long-term ADT; abiraterone added for high-risk node-positive or high-risk N0 disease (STAMPEDE)
Metastatic	ADT + ARPI (androgen receptor pathway inhibitor) ± docetaxel; RT to the prostate for low volume

- **Why surgery suits him:** young, gives pathological staging, and **removes the obstructing gland**, curing the LUTS. SPCG-4: surgical benefit is concentrated in men <65.
- Brachytherapy is relatively contraindicated with severe LUTS, a large gland or prior TURP (transurethral resection of the prostate) (retention, incontinence).

If biopsy is negative

- PSA 18 still needs explaining (prostatitis, large gland, missed anterior tumour): repeat MRI-targeted transperineal biopsy; %free PSA, PCA3, ConfirmMDx; close PSA follow-up.

Management of LUTS

- **Behavioural measures** (fluid timing, less caffeine) and an **α-blocker** (e.g. tamsulosin) at once; it does not alter PSA.
- **Avoid starting a 5ARI until cancer is excluded:** it halves PSA and masks cancer; if used later, record a baseline PSA and double readings after 12 months.
- Persistent storage symptoms with low PVR: add an antimuscarinic or β3-agonist (mirabegron).
- **Surgery** (TURP, HoLEP (holmium laser enucleation of the prostate)) for retention, refractory symptoms or complications, only once cancer is excluded. In locally advanced cancer with retention, a channel TURP relieves obstruction.
- **Also know:** pressure-flow study before surgery at age <50 with mainly voiding LUTS (EAU (European Association of Urology)).

```
PSA 18 + LUTS → history, DRE, urine culture → repeat PSA
      → mpMRI → targeted + systematic biopsy (transperineal)
   Cancer → risk group + staging → RP / RT + ADT / systemic
   Benign → explain PSA, re-biopsy if needed → BPH pathway
LUTS throughout: α-blocker; no 5ARI until cancer excluded
```

Diagram to draw: the flowchart above.

> **Key points to remember:**
> - PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
> - PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
> - Young fit man: RP treats both cancer and obstruction.
> - α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

Q2. Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder.

Type: LAQ (25 marks) (related: intestinal segments in 2023-Winter Paper 2, Q2; continent diversion and neobladder in Practice Set 3 Paper 4, Q2)

Uses of bowel

- Conduit (ileal, colon), continent cutaneous pouch (Indiana, Mainz), **orthotopic neobladder**, augmentation cystoplasty, ileal ureter, catheterisable channels (Mitrofanoff, Yang–Monti).

Principles of using bowel

1. Choice of segment

Segment	Metabolic effect	Use and drawbacks
Stomach	Hypochloraemic hypokalaemic alkalosis	Renal failure, short gut; haematuria–dysuria
Jejunum	Hyponatraemic, hyperkalaemic acidosis	Avoid (salt loss)
Ileum	**Hyperchloraemic metabolic acidosis**	Default: mobile, compliant; B12 and bile-salt loss
Colon	Hyperchloraemic acidosis, hypokalaemia	Easy tunnels; transverse colon outside radiation field

2. Surgical principles

- ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed. Mark the stoma site before surgery when a stoma is possible.
- Preserve the mesenteric arcade; tension-free, well-vascularised bowel anastomosis; close the mesenteric window.
- **Detubularisation:** opening bowel on its antimesenteric border and refolding it **abolishes coordinated peristaltic contractions**. A **sphere** gives the largest volume for its surface area; by Laplace's law (pressure ∝ wall tension/radius) a larger radius means lower pressure.
- **Ureterointestinal anastomosis:** spatulated, tension-free, mucosa to mucosa, stented. Refluxing (Bricker, Wallace; stricture about 3–6%) versus antireflux tunnels (more strictures). Obstruction, not low-pressure reflux, damages kidneys.

3. Patient selection for continent diversion

- Creatinine <2 mg/dL or GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; urine osmolality ≥600 mOsm/kg after water deprivation; normal liver function; ability to self-catheterise.

4. Complications to plan for

- **Metabolic:** ammonium chloride absorption → hyperchloraemic acidosis (treat with bicarbonate or citrate); hypokalaemia; B12 deficiency (after 5–7 years); osteomalacia; drug reabsorption.
- Mucus, stones (mucus, staples), bacteriuria, ureterointestinal stricture, pouch rupture, bowel obstruction, secondary malignancy (ureterosigmoidostomy, augments).

Orthotopic neobladder

Indication: after radical cystectomy, when the urethra is tumour-free, to allow voiding per urethra without a stoma.

Principles: (1) reliable outlet: intact rhabdosphincter and tumour-free urethra; (2) low pressure; (3) capacity ≥500 mL when mature (44 cm double-folded ileum: <200 mL early,

500–600 mL by 1 year); (4) unobstructed upper tracts.

Contraindications: positive urethral margin (frozen section), urethral stricture, poor renal or liver function, inability to catheterise. Relative: prostatic stromal invasion, bladder-neck or anterior vaginal tumour, prior radiation, frailty. Age and obesity alone are not bars.

Techniques compared

Design	Segment and shape	Ureters	Advantage	Drawback
Studer	40–44 cm ileum, double-folded U + afferent limb	Refluxing, isoperistaltic limb	Simple, no staples, short ureters; stricture <3%	Reflux on cystogram (benign)
Hautmann	About 70 cm ileum, W	Refluxing (since 1997)	Largest early capacity, early continence	More late retention, more absorption
T pouch	44 cm + 15 cm T limb	Flap-valve antireflux	Accepts dilated or short ureters	More reoperations than Studer
Abol-Enein	40 cm ileum, W	Serous-lined tunnel	Durable antireflux	Needs long, non-dilated ureters
Camey II	65 cm ileum, U	Le Duc tunnel	Early detubularised design	Le Duc strictures 7–29%
Hemi-Kock	61 cm ileum	Intussuscepted nipple	Good early results	Nipple stenosis, staple stones
Padua (VIP)	About 42 cm ileum, funnel	Le Duc	Anatomical funnel; robotic	—
Mainz III	10–15 cm caecum + 20–30 cm ileum	Submucosal tunnel	Good compliance	Loses ileocaecal valve
Sigmoid (Reddy)	35 cm sigmoid, U	Tunnelled	Spares ileum (short bowel)	Poor compliance, urgency

- **Hautmann vs Studer:** larger initial capacity gives earlier continence but more late retention and absorption.
- **Studer steps:** isolate ileum 15–20 cm proximal to the ileocaecal valve; open distal 40–44 cm antimesenterically; sew into a U and fold into a sphere; ureters end-to-side to the afferent limb; dependent point to urethra.
- **USC (University of Southern California) RCT (randomised controlled trial):** Studer vs T pouch, no difference in renal function at 3 years; T pouch needed more secondary procedures.

Outcomes

Outcome	Result
Day continence	80–90%
Night incontinence	20–50% (no sphincter reflex, mucosal water secretion)
Retention needing CIC (clean intermittent catheterisation)	Men 4–10%; women 20–30%
Urethral recurrence (men)	About 7% at 5 years

- **Also know:** robotic intracorporeal neobladder comparable to open; uptake is low despite 80–90% of men being eligible; lifelong follow-up of electrolytes, B12, upper tracts.

Diagram to draw: Studer pouch with afferent limb, ureters and urethral anastomosis.

Key points to remember:

- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemia; stomach: alkalosis.
- Detubularise and make a sphere; ≥500 mL; GFR >35 for continent diversion.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Q3a. Role of UDS before prostate surgery

Type: SAQ (10 marks)

Why consider UDS (urodynamic study)

- Symptoms, flow rate, PVR (post-void residual), prostate volume and cystoscopy **cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**: low Qmax (maximum flow rate) occurs in both. DUA is found in 9–48% of men with LUTS (lower urinary tract symptoms) studied.
- Surgery helps obstruction, not a weak bladder. **PFS (pressure–flow study) is the gold standard** for BOO.

Components

- **Filling cystometry:** sensation, compliance, **DO (detrusor overactivity)**.
- **PFS:** detrusor pressure at Qmax plotted against flow (ICS (International Continence Society) nomogram).

Index	Formula	Interpretation
BOOI (bladder outlet obstruction index)	PdetQmax − 2Qmax	>40 obstructed; 20–40 equivocal; <20 unobstructed
BCI (bladder contractility index)	PdetQmax + 5Qmax	>150 strong; 100–150 normal; <100 weak

- High pressure, low flow = BOO; low pressure, low flow = DUA.
- **Videourodynamics** adds anatomy (diverticula, reflux).

Indications before BPH (benign prostatic hyperplasia) surgery (EAU (European Association of Urology))

- Previous failed invasive treatment.
- Cannot void >150 mL (unreliable flow).
- PVR >300 mL.
- Age >80 years with mainly voiding LUTS.
- Age <50 years with mainly voiding LUTS.
- **Also:** suspected neurogenic bladder, prior pelvic surgery or radiation, Qmax >10 mL/s (AUA (American Urological Association): diagnostic uncertainty).

How results change management

- **Unobstructed or DUA:** poorer surgical results; counsel, consider CIC (clean intermittent catheterisation).
- **DO with BOO:** storage symptoms persist in about one-third after TURP (transurethral resection of the prostate); warn the patient.
- **Low compliance:** the only "absolute" urodynamic indication to relieve BOO (protects upper tracts).

Evidence and limitations

- Not routine. **UPSTREAM RCT (randomised controlled trial):** routine UDS did not change IPSS (International Prostate Symptom Score) or surgery rates.
- Invasive: UTI (urinary tract infection), discomfort, cost. Before **radical prostatectomy**, not routine.

Diagram to draw: ICS nomogram with obstructed, equivocal and unobstructed zones.

Key points to remember:

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

Q3b. Zolendronic acid

Type: SAQ (10 marks) (related: bone metastases in Practice Set 3 Paper 2, Q2)

Read as zoledronic acid.

Introduction

- A potent IV (intravenous) **nitrogen-containing bisphosphonate**.

Mechanism

- Binds hydroxyapatite at sites of active bone turnover and is taken up by **osteoclasts**.
- Inhibits **farnesyl pyrophosphate synthase** (mevalonate pathway), causing osteoclast dysfunction and apoptosis, so resorption falls.
- Breaks the "vicious cycle": tumour cytokines activate osteoclasts, and resorbed bone releases growth factors that feed the tumour.
- Excreted unchanged by the kidneys.

Indications and doses

Indication	Dose
SRE (skeletal-related event) prevention, mCRPC (metastatic castration-resistant prostate cancer) and other bone metastases (RCC (renal cell carcinoma), bladder)	**4 mg IV over ≥15 min every 4 weeks**, adjusted for creatinine clearance
ADT (androgen deprivation therapy) bone loss (by DEXA (dual-energy X-ray absorptiometry) and FRAX)	4–5 mg IV yearly
Hypercalcaemia of malignancy	4 mg IV with saline hydration

- 12-weekly dosing after the first doses is non-inferior to 4-weekly (CALGB 70604).
- **Not indicated at SRE doses in mHSPC (metastatic hormone-sensitive prostate cancer):** no benefit (CALGB 90202, STAMPEDE).

Evidence

- **Zometa 039 trial (mCRPC):** fewer SREs than placebo (about 33% vs 44%).
- **Denosumab vs zoledronate (1904 men, mCRPC):** time to first SRE 20.7 vs 17.1 months. **Neither improves survival.**

Precautions and monitoring

- Creatinine before every dose; avoid if creatinine clearance <30 mL/min.
- **Dental review before starting**; avoid invasive dental work during therapy.
- Calcium and vitamin D supplements.

Adverse effects

- **Acute-phase reaction** (fever, myalgia) after the first infusion.
- Nephrotoxicity, hypocalcaemia, hypophosphataemia.
- **ONJ (osteonecrosis of the jaw)**, atypical femoral fracture; ocular inflammation (rare).

Zoledronate vs denosumab

	Zoledronate	Denosumab
Target	Osteoclast enzyme	RANKL (receptor activator of nuclear factor-κB ligand)
Route	IV 4-weekly	120 mg SC (subcutaneous) 4-weekly
Renal adjustment	Yes	No

Diagram to draw: tumour–osteoclast vicious cycle with zoledronate's site of action.

Key points to remember:

- 4 mg IV 4-weekly for SREs in mCRPC; adjust for creatinine clearance.
- No survival benefit; not for SREs in mHSPC.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).

Q3c. LHRH analogue

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 4, Q3b and 2017-Winter Paper 4, Q3c)

Physiology

- **LHRH (luteinising hormone-releasing hormone), also GnRH (gonadotrophin-releasing hormone)**, is a hypothalamic decapeptide released in **pulses**. It drives pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone); LH stimulates Leydig cells (about 95% of testosterone).
- **Continuous** exposure desensitises and down-regulates the pituitary receptor: the basis of agonist therapy.

Agonists

Drug	Formulation
Leuprolide	Depot 1, 3, 6 monthly
Goserelin	SC (subcutaneous) implant 1 or 3 monthly
Triptorelin	IM (intramuscular) depot
Histrelin	Implant, 1 year

- Substitution at **position 6** raises potency and half-life.

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Agonist → LH surge (flare) → receptor down-regulation
        → LH falls → castrate testosterone in 2–4 weeks
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- Survival **equals orchidectomy**. Castrate level **<50 ng/dL**; aim **<20 ng/dL** (orchidectomy median about 15).

Flare phenomenon

- LH rises up to **10-fold for 10–20 days**: bone pain, **spinal cord compression**, retention.
- **Prevent:** antiandrogen (bicalutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → antagonist or orchidectomy.

Antagonists

- Immediate receptor block, **no flare**, FSH falls.
- **Degarelix** (SC monthly): castrate by day 3 in >95%.
- **Relugolix** (oral). **HERO** vs leuprolide: sustained castration 97% vs 89%, faster recovery, fewer MACE (major adverse cardiovascular events).

Indications

- Metastatic hormone-sensitive prostate cancer (with ARPI (androgen receptor pathway inhibitor) or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** intermittent ADT (androgen deprivation therapy) acceptable; precocious puberty, endometriosis.

Side effects

- Hot flushes, loss of libido, ED (erectile dysfunction).

- **Osteoporosis:** DEXA (dual-energy X-ray absorptiometry), calcium, vitamin D, denosumab or zoledronic acid.
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%); slow testosterone recovery.

Diagram to draw: hypothalamic–pituitary–testis axis showing agonist and antagonist sites.

Key points to remember:

- Agonist: surge, then down-regulation; castrate in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; antiandrogen 21–28 days.
- Castrate <50 ng/dL; aim <20 ng/dL.
- Antagonist: no flare; preferred in cord compression.

Q3d. Expectant treatment of CA prostate

Type: SAQ (10 marks)

Rationale

- Overdiagnosis: autopsy prevalence of microfocal cancer roughly equals age as a percentage; lifetime risk of death is about 3%.
- **Grade group 1 (Gleason 6) has virtually no metastatic potential.** Deaths on observation come from higher grade missed at biopsy (about 30% upgraded at prostatectomy).

Two forms of expectant treatment

	AS (active surveillance)	WW (watchful waiting)
Patient	Fit for radical therapy, life expectancy ≥10 years	Limited life expectancy (age, comorbidity)
Intent	**Curative, deferred**	**Palliative**
Monitoring	PSA (prostate-specific antigen), DRE (digital rectal examination), MRI, biopsy	Infrequent PSA and review; no biopsy
Action	Radical treatment on reclassification	ADT (androgen deprivation therapy) or palliation for symptoms

Evidence

Trial	Result
SPCG-4 (pre-PSA)	Prostatectomy cut cancer death (19.6% vs 31.3%), mainly <65 years and higher grade
PIVOT	No significant mortality benefit; low-risk cancer death 2.7%
ProtecT (15 years)	Cancer death <3% in all arms; metastasis doubled on active monitoring (9.4% vs about 5%)

Selection for AS

- **Very low risk (Epstein):** grade group 1, ≤2 positive cores, no core >50%, PSA density <0.15.
- **Low risk:** grade group 1, PSA <10 ng/mL; all guidelines prefer AS.
- **Favourable intermediate risk** (small-volume grade group 2): selected cases.
- Caution: **BRCA2** carriers, high-volume disease.

AS protocol

- MRI and **confirmatory biopsy** within 1–3 years; PSA 6-monthly; DRE every 1–2 years; MRI every 18–24 months; biopsy every 3–5 years.
- **Triggers:** grade group ≥3 → treat; volume or PSA rise → MRI or biopsy first.

Outcomes

- Cancer mortality **0.5–5% at 15 years** by entry criteria (Johns Hopkins vs Toronto); about **25% treated by 5 years**.

Watchful waiting

- Switch AS to WW when life expectancy falls below about 5–10 years.
- **Also know:** genomic tests refine selection; 5ARI (5α-reductase inhibitor) has no proven role.

Diagram to draw: AS flowchart: diagnosis → MRI + confirmatory biopsy → surveillance → trigger → treatment.

> **Key points to remember:**
>
> - AS is deferred cure; WW is palliation.
> - Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
> - ProtecT: no mortality difference at 15 years; more metastases on monitoring.
> - About 25% leave AS by 5 years.

Q3e. Molecular staging of CA prostate

Type: SAQ (10 marks) (related: prognosis in 2019-Summer Paper 4, Q3e)

Definition and rationale

- Clinical staging (DRE (digital rectal examination), PSA (prostate-specific antigen), grade, imaging) misses occult spread and cannot fully predict behaviour.
- **Molecular staging** uses DNA, RNA or protein markers in tissue, blood or nodes to find hidden disease and predict outcome.

1. Detection of occult spread

- **RT-PCR (reverse transcription polymerase chain reaction)** for PSA or PSMA (prostate-specific membrane antigen) mRNA in blood, marrow or "negative" nodes: detects micrometastases, but inconsistent, so not used clinically.
- **CTCs (circulating tumour cells):** counts are prognostic in metastatic disease; **AR-V7** (splice variant lacking the ligand-binding domain) predicts resistance to ARPIs (androgen receptor pathway inhibitors).
- **ctDNA (circulating tumour DNA)** liquid biopsy: DNA-repair mutations; clonal haematopoiesis gives false calls.

2. Tissue genomic classifiers

Test	Basis	Predicts
Decipher	22-RNA classifier	Metastasis, cancer death; post-prostatectomy radiotherapy decision
Oncotype DX GPS (Genomic Prostate Score)	17 genes (12 cancer, 5 reference) by RT-PCR	Adverse pathology; AS (active surveillance) choice
Prolaris	31 cell-cycle progression genes	Progression, cancer death

- **Also know:** PORTOS (24-gene radiotherapy response score); ConfirmMDx (methylation, after negative biopsy).

3. Genetic markers

- **Germline:** BRCA2 (aggressive disease), BRCA1, ATM, CHEK2, Lynch genes, HOXB13 (risk, not aggressiveness); test in metastatic or high-risk disease.
- **Somatic:** PTEN loss, TMPRSS2:ERG fusion, homologous recombination repair defects (guide PARP (poly ADP-ribose polymerase) inhibitors).

4. Molecular imaging

- **PSMA PET (positron emission tomography):** more sensitive and specific than CT and bone scan for nodes and metastases (proPSMA), but cannot exclude micrometastases.

Clinical use

- Refines **AS vs treatment** and post-operative radiotherapy decisions; guides targeted therapy.

Limitations

- Mostly retrospective; **multifocality and heterogeneity** (the sampled focus may not be the worst); cost; added value still being tested (G-MAJOR trial).

Diagram to draw: clinical stage vs molecular stage (tissue, blood, nodes, imaging).

Key points to remember:

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data.

Q3f. CRE & BEP

Type: SAQ (10 marks)

"CRE" is read as carbapenem-resistant Enterobacterales, and BEP as bleomycin, etoposide and cisplatin chemotherapy for germ cell tumours; the two are answered separately.

Part A: CRE (carbapenem-resistant Enterobacterales)

- **Definition:** Enterobacterales (Klebsiella pneumoniae, Escherichia coli, Enterobacter) resistant to at least one carbapenem, or producing a carbapenemase.

Mechanism	Example
Class A carbapenemase	KPC (Klebsiella pneumoniae carbapenemase)
Class B metallo-β-lactamase	**NDM (New Delhi metallo-β-lactamase)**: common in India
Class D	OXA-48
Non-enzymatic	ESBL (extended-spectrum β-lactamase) or AmpC plus porin loss

- **Urological risk:** catheters, stents, prolonged stay, prior carbapenem or fluoroquinolone use, transrectal biopsy, ICU (intensive care unit) care.
- **Treatment** (by culture and carbapenemase type; ID (infectious disease) input):
  - KPC, OXA-48: ceftazidime–avibactam; meropenem–vaborbactam or imipenem–relebactam for KPC.
  - **NDM:** ceftazidime–avibactam plus aztreonam, or cefiderocol.
  - Cystitis: single-dose aminoglycoside, fosfomycin; polymyxins as last resort. Avoid tigecycline for UTI (urinary tract infection) (low urine levels).
- **Prevention:** antibiotic stewardship, contact precautions, rectal screening of high-risk patients, early catheter removal, transperineal rather than transrectal biopsy.

Part B: BEP (bleomycin, etoposide, cisplatin)

- Standard first-line chemotherapy for metastatic **GCT (germ cell tumour)**; replaced PVB (less neuromuscular toxicity).
- **Usual 21-day cycle:** cisplatin 20 mg/m² days 1–5, etoposide 100 mg/m² days 1–5, bleomycin 30 units days 1, 8, 15.

IGCCCG (International Germ Cell Cancer Collaborative Group) risk	Regimen
Good	BEP ×3 or EP ×4
Intermediate/poor	BEP ×4 (VIP ×4 if lungs poor)
Stage I NSGCT (non-seminomatous GCT), high risk	BEP ×1 adjuvant

- **Toxicity:** bleomycin lung fibrosis (keep FiO_2 low at later RPLND (retroperitoneal lymph node dissection)) and Raynaud phenomenon; cisplatin nephrotoxicity, ototoxicity, neuropathy; myelosuppression; infertility (bank sperm); late cardiovascular disease and second cancers.

Diagram to draw: BEP 21-day cycle timeline.

> **Key points to remember:**
> - CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
> - BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
> - Bleomycin lung: low FiO_2 and restricted fluids at surgery.

Rapid Revision Sheet – Practice Set 14 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief. (LAQ)

- PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
- PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
- Young fit man: RP treats both cancer and obstruction.
- α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

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- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemia; stomach: alkalosis.
- Detubularise and make a sphere; ≥500 mL; GFR >35 for continent diversion.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Q3a. Role of UDS before prostate surgery (SAQ)

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

Q3b. Zolendronic acid (SAQ)

- 4 mg IV 4-weekly for SREs in mCRPC; adjust for creatinine clearance.
- No survival benefit; not for SREs in mHSPC.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).

Q3c. LHRH analogue (SAQ)

- Agonist: surge, then down-regulation; castrate in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; antiandrogen 21–28 days.
- Castrate <50 ng/dL; aim <20 ng/dL.
- Antagonist: no flare; preferred in cord compression.

Q3d. Expectant treatment of CA prostate (SAQ)

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Q3e. Molecular staging of CA prostate (SAQ)

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data.

Q3f. CRE & BEP (SAQ)

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.

- Bleomycin lung: low FiO_2 and restricted fluids at surgery.